SUMMARY (cont.)

- If natural frequencies are written as integral multiples of fundamental frequency, then the frequencies are said to be in harmonics. Thus, the first harmonic is $v_1 = v_1$, (the fundamental frequency is called first harmonics), the second harmonics is $v_2 = 2\ v_1$, the third harmonics is $v_3 = 3\ v_1$, and so on.
- *Loudness of sound* is defined as *"the degree of sensation of sound produced in the ear or the perception of sound by the listener".*
- *The intensity of sound* is defined as *"the sound power transmitted per unit area placed normal to the propagation of sound wave ".*
- Sound intensity level, $\Delta L = 10 \log_{10} \frac{I_1}{I_0}$ *decibel* .
- A closed organ pipe has only odd harmonics and the corresponding frequency of the nth harmonic is $f_n = (2n + 1) f_1$.
- In a closed organ pipe the frequencies of harmonics are in the ratio

  $f_1 : f_2 : f_3 : f_4 : ... = 1 : 3 : 5 : 7 : ...$
- The open organ pipe has all harmonics and frequency of the nth harmonic is $f_n = n f_1$.
- In the open organ pipe the frequencies of harmonics are in the ratio

  $f_1 : f_2 : f_3 : f_4 : ... = 1 : 2 : 3 : 4 : ...$
- Whenever there is a relative motion between the source of sound and the listener, the frequency of the sound observed by the listener is different from the frequency produced by the source. This is known as Doppler effect.

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